Ling 285

Review for Exam 2 (HW6, Everyone gets 100)

Answers will be gone through at Tue 11/10 Review

1. Why do multiple mass systems have multiple frequencies?

2. How many modes does a 3-mass system has?
3. Which Mode of the 3-mass system has the highest frequency? Why?

4. If you increase mass 2, what happens to the frequencies of each of the modes?

5. What do you get when you have an infinite number of masses and springs?
6. This is Mode 5 of a string. Circle the areas where there is most springiness.



7. The string above is fixed on both sides. If mode 1 has frequency 200 Hz, what frequency does Mode 5 have?
8. Why are the frequencies of fixed-fixed vibratory systems spaced the way they are?

9. Draw the spectrum for the string above up to mode 6.

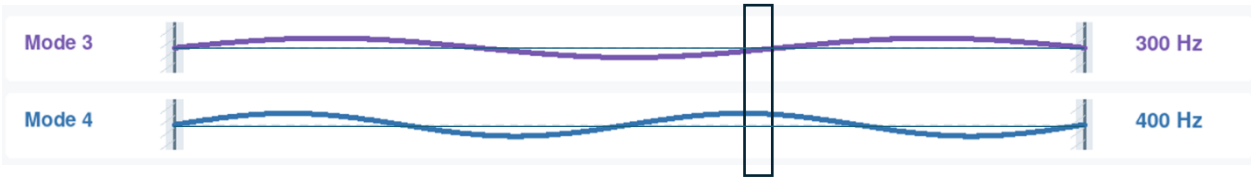
10. If vocal folds have their base frequency, F0, at 150, draw the spectrum for the first 5 harmonics.

11. This picture shows mode 4 of a building. Circle the areas where there is the most springiness. 

12. What in speech is this building analogous to?

13. Why does this analogy hold?

14. if we increase the stiffness and mass of the string at the location in box, what happened to Mode 3 and Mode 4's frequencies and why?



15. 

If we increase the mass and stiffness of the building at its very top, what would happen to the frequency of vibration of mode 4, and why?

16. Why are the formants of fixed-free system 1,3,5 etc. multiples of the base frequency?

17. What aspect of speech has the properties in 16?

18. Describe a spectrogram.

19. If I constrict the vocal tract somewhere and I raise the formant a great deal, what must be true of air motion vs. springiness at this location?

20. How many hidden layers does this NN have?

```
net = nn.Sequential(
    nn.Linear(2, 3),
    nn.Linear(3, 2),
    nn.Linear(2, 4),
    nn.Linear(4, 5)
)
```

21. Write the code for an NN with an additional hidden layer right before the output layer with 3 neurons, and adjust the other layers.